Regression

1. Introduction to Regression

Regression is one of the most fundamental concepts in statistics and machine learning. It is a supervised learning algorithm used for predicting continuous numerical values. The basic goal of regression is to understand the relationship between one dependent (target) variable and one or more independent (input) variables.

For example, if you want to predict a student's exam score based on the number of hours studied, regression helps you build a mathematical relationship between the two.

Formally, regression analysis allows us to model the expected value of a dependent variable as a function of independent variables. The mathematical model obtained can then be used for prediction, forecasting, and inference.

2. Target Attribute (Dependent Variable)

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The **target attribute**, also known as the **dependent variable**, **output variable**, or **response variable**, is the variable that we aim to predict or explain.

It represents the **outcome of interest** whose value depends on other variables (predictors).

In mathematical terms, if we denote the target variable as Y, and the predictors as X1,X2,X3,.....,Xn,

the regression model seeks to express Y as a function of Xi.

$Y=f(X1,X2,X3,...,Xn)+\epsilon$

Here, ϵ represents the **error term**, accounting for the random variability or noise not explained by the predictors.

Examples of Target Attributes:

- Predicting **house prices** → Target: Price
- Predicting **sales revenue** → Target: Revenue
- Predicting **temperature** → Target: Temperature
- Predicting **stock price** → Target: Price of a stock

The target attribute is always numerical or continuous in regression problems, unlike classification tasks, where the target variable is categorical (e.g., "Yes"/"No" or "High"/"Low").

3. Descriptive Attribute (Independent Variable)

The **descriptive attributes** (or **independent variables**) are the **input features** used to describe, explain, or predict the target attribute. They are also known as **predictors**, **explanatory variables**, or **regressor variables**.

Descriptive attributes are measurable characteristics that influence the dependent variable.

For example:

- When predicting house prices, the descriptive attributes might include **area, number of rooms, location, and age of the house**.
- When predicting student grades, they may include **study hours, attendance percentage, and assignment scores**.

These attributes are used as inputs to the regression model to compute the output (target variable).

Mathematically, for a simple linear regression:

$Y=\beta0+\beta1X+\epsilon$

Here, X is the descriptive attribute.

In multiple regression:

$Y=\beta0+\beta1X1+\beta2X2+...+\betanXn+\epsilon$

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4. Response Variable and Regressor Variable

Response Variable

The **response variable** is another term for the **target attribute**. It represents the quantity that responds to changes in the input variables.

For instance, in a study relating rainfall and crop yield:

- Response variable → Crop yield (dependent on rainfall)
- Predictor variable → Rainfall

Thus, the **response variable** measures the effect or outcome.

Regressor Variable

The **regressor variables** are the **independent variables** that help in predicting or explaining the response.
They are the input features or explanatory variables whose variations lead to variations in the response.

In the model:

$Y=f(x)+\epsilon$

Y=Response variable

X=Regressor variable

Both terms are closely linked:

Response variable = Target/Dependent variable

Regressor variable = Predictor/Independent variable

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5. Parametrized Function in Regression

A **parametrized function** is a mathematical representation that defines how input variables (regressors) are related to the output (response variable) through certain parameters.

In regression, we assume a **functional form** (usually linear or nonlinear) with **parameters** that need to be estimated using data.

For example, in **linear regression**, the parametrized function is:

$Y=\beta0+\beta1X+\epsilon$

Here:

$\beta0$= Intercept (parameter)

$\beta1$=Slope (parameter)

ϵ= Random error

The goal of regression is to estimate the parameters $\beta0,\beta1,...\beta n$ such that the model fits the data well.

These parameters define the relationship between independent and dependent variables.

In nonlinear regression, the parametrized function might be exponential, logarithmic, or polynomial, such as:

$Y=\beta0+\beta1X+\beta2X2+\epsilon$

6. Interpolation and Extrapolation Purposes

Regression models are used for two key purposes: interpolation and extrapolation

6.1 Interpolation

Interpolation means predicting values within the range of observed data points.

For instance, if you have data on house prices for sizes between 1000 and 2000 square feet, using regression to predict the price of a 1500 sq.ft house is interpolation — because 1500 lies within the known data range.

Interpolation is generally reliable since it assumes the same data relationship holds within the existing range.

Mathematically, if we have data points:

(X1,Y1),(X2,Y2),.....,(Xn,Yn)

and we estimate Y for a new X such that Xi<X<Xi+1,it's interpolation.

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6.2 Extrapolation

Extrapolation involves predicting values **beyond the range** of observed data.

If the same house price model is used to predict the price of a 2500 sq.ft house, even though no such data exists in the sample, it is **extrapolation**.

Extrapolation is riskier because it assumes that the relationship observed in the data holds beyond the observed range, which may not always be true.

For instance, stock price trends might follow a linear pattern for a short period but deviate significantly later.

In short:

Interpolation → safer, within data range
Extrapolation → riskier, outside data range

7. Application Example: Investing in Stock

Regression is widely used in **financial analytics**, especially in **stock market prediction** and **investment analysis**.

The stock market involves numerous variables — such as interest rates, company earnings, GDP growth, and inflation — that influence stock prices. Regression models help understand and forecast how these variables affect returns.

7.1 Stock Price Prediction Example

Suppose we want to predict a company's stock price (Y) based on:

X1:Market index performance (e.g., NIFTY or S&P 500)

X2:Company earnings per share (EPS)

X3: Interest rate

A multiple linear regression model can be expressed as:

$Y=\beta0+\beta1X1+\beta2X2+\beta3X3+\epsilon$

Here, the coefficients $(\beta1,\beta2,\beta3)$measure the sensitivity of the stock price to changes in each predictor.

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7.2 Regression in Portfolio Management

Investors often use regression to understand:

- How sensitive a stock is to market fluctuations (Beta value)
- How much of the stock's return can be explained by market returns

For instance, in the Capital Asset Pricing Model (CAPM):

$Ri=Rf+\beta i(Rm-Rf)+\epsilon$

where:

- Ri = Return of stock *i*
- Rf = Risk-free rate
- Rm = Market return
- βi = Sensitivity (regression coefficient)

Here, regression estimates the Beta, showing how the stock moves relative to the market.

7.3 Practical Uses

- Predicting future stock returns
- Determining factors affecting stock volatility
- Assessing investment risk
- Forecasting market trends

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However, it's essential to remember that financial data is volatile, and regression models work best when relationships are stable and linear.

8. Linear Regression Algorithm

8.1 Definition

Linear Regression is the simplest form of regression that models the relationship between dependent and independent variables using a straight line.

It assumes that the dependent variable YYY can be expressed as a linear combination of the independent variables Xi:

$$Y=\beta0+\beta1X1+\beta2X2+...+\betanXn+\epsilon$$

Here:

- $\beta 0$: Intercept (value of Y when all X = 0)
- βi: Coefficients representing the impact of each independent variable
- ϵ: Random error (difference between actual and predicted values)

The goal is to find the best-fit line that minimizes the sum of squared errors (SSE) between predicted and actual values.

8.2 Working Principle

Linear regression works on the principle of Ordinary Least Squares (OLS).

OLS minimizes the sum of squared residuals:

$SSE=i=1\sum n(Yi-Yi\hat{})2$

where:

- Yi: Actual value
- Yi^: Predicted value from regression line

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The OLS algorithm computes the best-fitting line by solving for $\beta 0, \beta 1, ...\beta n$ that minimize this error.

8.3 Types of Linear Regression

1. **Simple Linear Regression**
    - One independent variable.
    - Example: Predicting sales based on advertising spend.
2. **Multiple Linear Regression**

- More than one independent variable.
- Example: Predicting house price based on area, bedrooms, and location.

$Y=\beta 0+\beta 1X1+\beta 2X2+...+\beta nXn+\epsilon$

8.4 Assumptions of Linear Regression

To ensure accurate results, linear regression assumes:

1. **Linearity** – The relationship between predictors and response is linear.
2. **Independence** – Observations are independent.
3. **Homoscedasticity** – Constant variance of errors.
4. **Normality** – Errors follow a normal distribution.
5. **No multicollinearity** – Independent variables are not highly correlated.

Violation of these assumptions can reduce the accuracy of the regression model.

8.5 Advantages

- Simple to understand and interpret.
- Works well when relationships are linear.
- Computationally efficient.
- Useful for both prediction and inference.

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8.6 Limitations

- Cannot capture non-linear relationships.
- Sensitive to outliers.
- Assumes constant variance and normality of errors.
- Poor performance on complex, high-dimensional data without feature engineering.

9. Prediction Using Linear Combination of Input Features

Linear regression predicts the target variable based on a linear combination of input features. This means each input feature contributes a weighted amount to the final prediction.

For instance:

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_n X_n$$

Each β_i acts as a weight showing how much influence the feature X_i has on the target variable.

Example:

Predicting house prices:

$$\text{Price} = 25000 + 100 \times (\text{Area}) + 5000 \times (\text{Bedrooms})$$

Here:

- Each additional square foot adds ₹100.
- Each additional bedroom adds ₹5000.
- The intercept ₹25,000 represents the base price.

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Thus, regression models help **quantify** the relationship between features and the target variable.

10. Conclusion

Regression plays a foundational role in **data analysis, forecasting, and machine learning**. It helps establish and quantify relationships between variables, making it essential for **decision-making, prediction, and trend analysis**.

Whether used to predict **house prices**, **student performance**, or **stock market trends**, regression provides a systematic way to model reality mathematically. Understanding core concepts like **target and descriptive attributes**, **response and regressor variables**, and **parametrized functions** enables analysts to build reliable models for both **interpolation and extrapolation**.

When applied responsibly — with awareness of its assumptions and limitations — regression remains one of the most powerful and interpretable tools for predictive analytics.